Mark Dodici

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RESEARCH INTERESTS

Transient phenomena in galactic nuclei; binary stars in dense clusters; hierarchical triples and few-body dynamics; intermediate-mass black holes; stellar tides; celestial mechanics; galactic dynamics

EDUCATION

University of Toronto 2022 – (expected Aug 2027)
Ph.D. candidate, Department of Astronomy & Astrophysics (UofT Astro)
Graduate researcher, Canadian Institute for Theoretical Astrophysics (CITA)
Visiting graduate student, Institute for Advanced Study, Winter 2026
Advisors: Professors Yanqin Wu & Scott Tremaine

Princeton University 2018 – 2022
A.B. Astrophysical Sciences, *magna cum laude*
Thesis advisors: Dr. Christopher Spalding & Professor Jeremy Goodman

FELLOWSHIPS, GRANTS, and AWARDS

Jui Lin (Allen) Yen Award 2026
UofT Astro, for outstanding student research
Lachlan Gilchrist Fellowship 2026
UofT, for excellence in physical sciences
Duncombe Student Research Prize 2026
AAS Division on Dynamical Astronomy
Carl Reinhardt Fellowship in Astronomy 2025
UofT Astro, for outstanding international students
Dunlap Institute Seed Funding Grants 2023, 2024
For development of outreach workshop series Coding the Cosmos *and* Age of the Universe*; see below*

PUBLICATIONS *3 first-author; 78 citations (33 first-author).*

FIRST AUTHOR

5. **M. Dodici**, 2026. Binary disruptions driven by massive disks around massive black holes. arXiv:2607.26205. Submitted to ApJ; received positive referee report. 24 pp.
4. **M. Dodici**, S. Tremaine, & Y. Wu, 2026. Dynamical evolution of stellar binaries in galactic centers. ApJ, 1000, 226. 13 pp.
3. **M. Dodici** & S. Tremaine, 2024. Studying binary formation under dynamical friction using Hill's problem. ApJ, 972, 193. 26 pp.

CONTRIBUTING

2. B. Hensley, C. Murray, & **M. Dodici**, 2022. Polycyclic aromatic hydrocarbons, anomalous microwave emission, and their connection to the cold neutral medium. ApJ, 929, 23. 9 pp.

ON EDUCATION

1. B. Lewis et al. (incl. **M. Dodici**), 2025. Improving undergraduate astronomy students' skills with research literature via accessible summaries: An exploratory case study with Astrobites-based reading assignments. Phys. Rev. Physics Education Research, 21, 010124. 17 pp.

IN PREPARATION

M. Dodici, Y. Wu, & S. Tremaine, in prep. Constraints on tidal dissipation rates from compact stellar triples.

M. Dodici & C. Ye, in prep. Models of 47 Tucanae with an intermediate-mass black hole.

TALKS

24 total, 6 invited (∗), 1 award-based (†). Unique topics noted.

AT CONFERENCES

24. AAS 249 — Salt Lake City, UT. Jan 2027
23. ∗ Galactic Nuclei Astrophysical Phenomena — Sexten, Italy. Jul 2026
    Dynamics of stellar binaries in the centers of galaxies
22. † AAS Division on Dynamical Astronomy 57 — Chicago, IL. Jun 2026
    Constraining tidal dissipation rates with compact stellar triples
21. CITA @ 40 / Kingston Meeting 2026 — Toronto, ON. May 2026
20. Traversing the Galactic Center in Space and Time / IAU Symposium 405 — Brno, Czechia. May 2026
19. MODEST–25 / IAU Symposium 398 — Seoul, South Korea. Jun 2025
    Stellar binaries in galactic nuclei often shrink to near-contact separations
18. AAS Division on Dynamical Astronomy 56 — Atlanta, GA. May 2025
17. AAS Division on Dynamical Astronomy 55 — Toronto, ON. May 2024
    Using Hill's problem to study binary formation under dynamical friction
16. AAS Division for Planetary Science 55 — San Antonio, TX. Oct 2023
    Breaking up a denser, primordial Neptunian scatter belt to supply the cold classical Kuiper Belt
15. CITA Planet Day — Toronto, ON. Aug 2023
14. Emerging Researchers in Exoplanet Science Symposium VIII — New Haven, CT. Jun 2023
    Finding the distribution of stellar obliquities for newly-formed planets in binary systems
13. Great Lakes Exoplanets Area Meeting — Columbus, OH. Nov 2022
12. AAS 240 — Pasadena, CA. Jun 2022
    A Trojan Horse for White Dwarfs: Co-orbital asteroid dynamics under stellar mass loss, radiative effects

AT INSTITUTIONS

11. ∗ Astronomy Lunch Seminar, University of California, Los Angeles — Los Angeles, CA. Dec 2026
10. Theory Talk, Northwestern University/CIERA — Evanston, IL. Nov 2026
9. Dynamics Group, University of Wisconsin, Madison — Madison, WI. Nov 2026
8. ∗ Tea Talk, Indiana University — Bloomington, IN. Nov 2026
7. ∗ APS Lunch Seminar, University of Colorado, Boulder — Boulder, CO. Oct 2026
6. Galactic Center Group, Max Planck Institute for Extraterrestrial Physics — Garching, Germany. May 2026
5. Rossi Group, Leiden Observatory — Leiden, Netherlands. May 2026
4. ∗ Dynamics Blackboard Talk, Institute for Advanced Study — Princeton, NJ. Mar 2026
3. ∗ Astrophysics Seminar, Institute for Advanced Study — Princeton, NJ. Feb 2026
2. Compact Objects Group, CITA — Toronto, ON. Feb 2024
1. Lunch Talk, UofT Astro — Toronto, ON. Jan 2024, Apr 2026

LEADERSHIP and SERVICE ROLES

Referee *ApJ, ApJL, OJA* — 2025 –

Coding the Cosmos, *Co-director (2024 –)* — 2023 –
Developed workshop series and educational materials for HS students

UofT Graduate Astronomy Students Association — 2022 –
Co-President (2024 – 2025)
Courses & Qualifying Exams Committee (2022 – 2024, 2025 –)

UofT Astro Department Jamboree, *Organizer & Emcee* — 2023 – 2025

UofT AstroTours, *Co-director (2023 – 2024)* 2022 – 2024

Public lecture/telescope events led by graduate students

Age of the Universe (HS workshop), *Co-organizer & Speaker* 2023

Progenitor workshop of Coding the Cosmos*; see above*

TEACHING ASSISTANT EXPERIENCE

Stars and Planets (AST221, Toronto) Fall 2024, 2025, 2026

The Sun and its Neighbours (AST101, Toronto) Fall 2022, 2023

Invited Guest Lecturer, Summer 2026

Introduction to Astrophysics (AST320, Toronto) Winter 2025

Stars and Galaxies (AST201, Toronto) Winter 2023, 2024

The Universe (AST203, Princeton) Winter 2021, 2022

OTHER OUTREACH

UofT AstroTours, *Speaker* 2026

Delivered public lecture, "Tides across the Universe," on tidal physics at all scales

Pursue STEM, *Workshop leader* 2025

HS program for Black students interested in physical sciences

Astrobites, *Author & Education Study co-author* 2023 – 2025

Summaries of astronomy literature for undergraduates; produced peer-reviewed paper on education

Eclipse 2024 at the Toronto Public Library, *Speaker* 2024

Series of workshops and lectures on solar eclipses

Cosmos from your Couch, *Video editor* 2023

Series of videos for general audiences featuring UofT Astro researchers

ComSciCon Canada, *Attendee* 2023

Competitive, nation-wide workshop for science communication

Princeton Research Day, *Award-winning presenter* 2021

Virtual talk competition, like Three-Minute Thesis *competitions*